

HAOLIN (ALEX) LIU

alexliu.bio | github.com/haolin122

EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING

Master of Science in Financial Engineering, GPA: 3.56/4.0

Brooklyn, NY

05/27

UNIVERSITY OF WASHINGTON (SEATTLE CAMPUS)

Bachelor of Science Degree in Computational Finance and Risk Management, GPA: 3.72/4.0

Seattle, WA

06/25

PROGRAMMING / TECHNICAL SKILLS / CERTIFICATIONS

Programming

- Programming Skills: Java, Python, SQL, R, Matlab

COURSEWORK HIGHLIGHTS

- **Mathematics & Stochastic Modeling:** Quantitative Methods in Finance
- **Derivatives & Risk Management:** Introduction to Derivative Securities, Financial Risk Management
- **Statistics & Machine Learning:** Financial Econometrics, Machine Learning in Financial Engineering
- **Programming:** Financial Computing in C++

ACADEMIC PROJECTS

GARCH OPTION PRICING & IMPLIED VOLATILITY (S&P 500), Brooklyn, NY

11/25 - 12/25

Group Member

- Fitted GARCH(1,1) on 5 years of S&P 500 data to forecast time-varying volatility; simulated index paths via Monte Carlo to price European options as discounted expected payoffs
- Constructed 3D implied volatility surface across strikes and maturities by inverting Black-Scholes; found BS overprices deep OTM calls by up to 8.6% versus GARCH-based pricing

FX AMBIGUITY & SAFE-HAVEN RESERVE ALLOCATION, Brooklyn, NY

09/25 - 11/25

Group Member

- Constructed a quarterly FX ambiguity index from daily spot returns of 9 major currency pairs using the Asano (2025) methodology; merged with IMF COFER reserve data to test whether ambiguity shocks shift central-bank allocations toward USD and CHF
- Ran OLS regression and correlation analysis on 37 quarterly observations (2016-2025); found a weak negative relationship ($r = -0.36$), suggesting central banks do not increase safe-haven holdings during elevated FX ambiguity, contrasting with private-market carry-trade behavior

EARNINGS SURPRISE EVENT STUDY ON RUSSELL 3000 (C++), Brooklyn, NY

03/26 - 05/26

Group Member

- Built modular OOP framework conducting sector-neutral event study on ~2,500 Russell 3000 earnings announcements; grouped stocks into Beat/Meet/Miss by surprise %, retrieving adjusted close prices through libcurl-based EOD API calls parallelized with std::async
- Implemented bootstrap engine (40 iterations x 30 stocks resampled with replacement) computing $E[AAR]$, $E[CAAR]$, t-statistics, and 95% CIs; rendered CAAR curves with gnuplot, finding Beat stocks exhibit positive post-announcement momentum while Miss stocks produce the strongest negative abnormal returns

EXPERIENCE

CHINA SECURITIES CO., LTD. (CSC), Beijing, CN

07/26 - 09/26

FinTech Intern, Information Technology Department

- Owned end-to-end backend development of an AI-assisted narrative generation feature for the firm's anti-money laundering (AML) system (Java, SQL, Spring Boot, MyBatis, MySQL): designed a submit-task / async-invoke / poll-result API chain that queries 13 customer profile fields from data warehouse parameter tables (TDSQL), merges them with relationship managers' due-diligence inputs, and calls the firm's internal LLM agent platform to generate customer due-diligence narratives

DONGHAI SECURITIES, Beijing, CN

06/24 - 08/24

Investment Department Intern

- Performed macro/meso/micro market analysis; applied logistic regression on 10+ years of data to forecast CSI All Share Index trends, outperforming baseline benchmarks in predicting rising vs. falling curves

GREEN FUND, Tianjin, CN

06/23 - 09/23

Summer Intern, Investment Department

- Collected data regarding yield rate of mid- and long-term bond funds in China for most recent 5 years, summarized and analyzed data, and found cumulative yield rate achieves 19.4% and annual compound yield rate at 3.8%